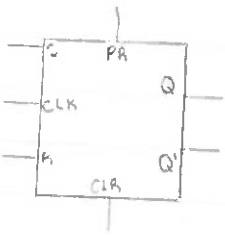
Asynchronous inputs

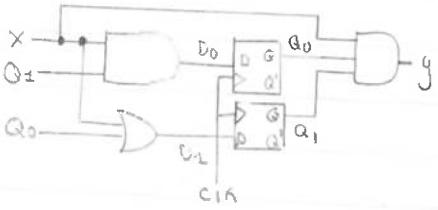
Def: inputs that operate independently of the synchronous inputs & clock.

- used to set the flip-flop to 1 or 0 state at any time.



Flip-flop state

- Behavior of synchronous sequential circuit can be determined from inputs, outputs & Flip flop state.



State equations:

Specifies the next state as a function of the present state & inputs.

State table:

- Sequence of outputs, inputs, & flip flop states
- present state indicates current value of flip flops
- next state indicates state after next clock edge
- output is output value on current clock edge

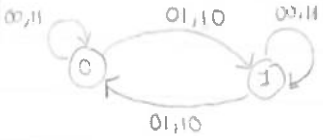
Present State	next state x=0	next state x=1	output x=0	output x=1
0 0	00	10	0	0
0 1	10	10	0	0
1 0	00	11	0	0
1 1	10	11	0	1

$Q_0(t+1) = D_0(t) = X(t)\,Q_1(t)$

$Q_1(t+1) = D_1(t) = X(t) + Q_0(t)$

State diagram

- Circles indicate next state
- Arrows point to next state



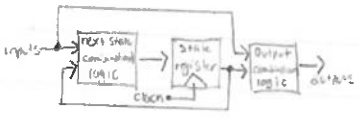
ex

present state	inputs		next state
A	x	y	A
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

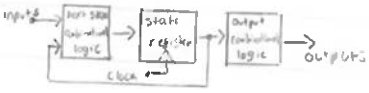
Mealy & Moore models

model of sequential circuits: inputs, outputs, internal states

mealy machine



moore machine



State reduction

- procedures for reducing the number of states in a state table, while keeping the external input-output requirements unchanged

present state	next state x=0	next state x=1	output x=0	output x=1
a	i	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
-> e	a	f	0	1
f	g	f	0	1
-> g	a	f	0	1

present state	next state x=0	next state x=1	output x=0	output x=1
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
-> d	c	f	0	1
e	a	f	0	1
-> f	e	f	0	1

a	a	b	0 0
b	c	d	0 0
c	a	d	0 0
d	e	d	0 1
e	a	d	0 1

